

# Staff & Senior UX Research Portfolio Repository: Comprehensive Study Catalog

**Author:** [nasun@google.com](mailto:nasun@google.com) | **Total Archived Studies:** 61 Studies (2019 – 2026)

**Corpus:** Google Research Archive & UX Knowledge Graph (UXKG)

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## Executive Portfolio Summary & Strategic Positioning

This document serves as the master portfolio reference and comprehensive study archive for **nasun@ (Staff / Senior UX Researcher)**. It synthesizes **61 archived research studies** across 4 major Google product pillars, documenting technical complexity, methodological rigor, and multi-year product impact.

*Core Narrative Archetype: The Full-Stack Platform & Intelligence Researcher*

Most UX researchers specialize strictly in either consumer frontend interfaces or enterprise management consoles. This portfolio demonstrates rare **full-stack technical depth** spanning:

1. **Headless Developer & Data Infrastructure:** Indexing pipelines, C++ ORMs, GoogleSQL, and multi-year quantitative CSAT benchmarking.
  2. **Platform Ecosystems & Extensibility:** 3-sided marketplaces (Admins, End-Users, 3P Developers), Chat bots, and workflow automation.
  3. **Enterprise Security, Identity & Governance:** IAM administration, RBAC, Unified Roster, guest lifecycle management, and agentic AI identity.
  4. **Frontier Generative AI & Multimodal Creation:** Direct-manipulation image generation, cognitive onboarding for low-AI-literacy creators, and AI memory models.
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## Part I: Strategic Analysis — Is Maintaining a Personal UXR Portfolio / Website Worthwhile?

### 1. The Short Answer

**Yes, but its format and purpose change dramatically as you reach Senior, Staff, and Principal levels.** At junior/mid levels, portfolios focus on *tactical execution* (e.g., "I ran 8 usability sessions and made wireframe recommendations"). At **Senior/Staff levels**, hiring managers, executives, and promotion committees look for **strategic influence, technical domain breadth, portfolio-level problem framing, and organizational impact**.

### 2. High-Value Benefits of a Senior/Staff Portfolio

- **Executive Pitch & Promotion Tool:** An organized portfolio enables you to quickly assemble promotion packets, lead executive review deep-dives, and demonstrate multi-year strategic arc.

- **Cross-PA Mobility & Leadership Visibility:** When considering transfers across product areas (e.g., Core Infra to GenAI), having a centralized dossier immediately proves your domain versatility.
- **Thought Leadership & Personal Brand:** A personal site or internal g3doc/Sites page establishes you as an authority on complex subjects (e.g., "Enterprise IAM UX", "Developer Platform Extensibility", "Low-AI-Literacy GenAI Onboarding").
- **Intellectual Asset Management:** With 60+ studies across 7 years, maintaining a structured index prevents past insights from being lost during team transitions or reorgs.

### 3. Recommendations: How to Structure a Senior/Staff UXR Portfolio Website / Doc

1. **Frame Projects by Strategic Narrative, Not Chronology:** Group your work into thematic pillars (the 4 pillars below) rather than a flat chronological list.
2. **Emphasize the "Problem Behind the Problem":** Highlight how you reframed ambiguous business problems (e.g., shifting Unified Roster from end-user sharing to an administrative visibility cockpit).
3. **Showcase Methodological Versatility:** Highlight sophisticated quantitative methods (MaxDiff, TURF, longitudinal Welch's t-tests, log telemetry triangulation) alongside deep ethnographic/foundational field research.
4. **Demonstrate Ecosystem Thinking:** Show how your research accounted for multi-stakeholder dynamics (e.g., how developer API friction affects IT admin adoption and end-user engagement).
5. **Confidentiality & Sanitization:** If maintaining an *external* portfolio website, always scrub proprietary metrics, code symbols, unannounced codenames, and PII. Frame case studies around generalized archetypes, methodologies, and UX frameworks. For *internal* portfolios (e.g., g3doc or Google Sites), full internal links and data are ideal.

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## Part II: The 4 Portfolio Pillars & Complete Study Catalog

### Theme 1: Enterprise Identity, Security Governance & Administration (IAM)

**Focus:** Identity & Access Management (IAM), Role-Based Access Control (RBAC), Group administration, Unified Roster, Guest Pass / External Collaboration, and Agentic AI Identity.

**Total Studies:** 14

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#### 1. Guest Pass OU

- **Date / Status:** 2025-10-01 | **Team:** Workspace Platform
- **Deliverables:** [Deliverable](#)

**Summary & Research Context:** Admins recognize the Guest Organizational Unit (OU) as a security-focused tool for managing temporary access for external users, though many lack

specific use cases for its application. Most participants do not find the Guest OU distracting as it often goes unnoticed during routine tasks. There is significant overlap in how admins perceive OUs and Groups, highlighting a need for better differentiation and user guidance within the Admin Console.

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## *2. Groups Admins' Challenges and Opportunities (go/groups-admin-uxr)*

- **Date / Status:** 2025-12-11 | **Team:** Workspace Platform
- **Deliverables:** [Groups Admins' Challenges and Opportunities \(go/groups-admin-uxr\)](#)

**Summary & Research Context:** Admins face significant challenges with Google Groups management due to a lack of visibility, proliferation, and tedious manual updates. The proposed Unified Roster concept, including non-email-addressable 'Plain Groups', is strongly validated but requires better governance and administrative controls to be successful.

### **Key Insights & Takeaways:**

- Top pain points of group administration: The absence of centralized visibility, reporting, and lifecycle management is causing widespread Group Proliferation and forcing administrators to actively suppress the adoption of Groups as a core collaboration primitive. The success of Unified Rosters depends on immediately addressing these governance deficits, while addressing two prerequisites.
  - Unified Roster Concept Exploration: Admins strongly validate the "Plain Group" (non-email-addressable) concept for security, but require a unified, centralized management portal for Group policies (creation, naming) and real-time usage reporting (roles, permissions, connected services).
  - While admins acknowledge the end-user benefits of the unified roster, they have significant security concerns. The current strategy's focus on end-users does not fully address core admin challenges like visibility and proliferation, and future milestones where every chat room becomes a group could exacerbate these issues
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## *3. Unified Roster Admin Study*

- **Date / Status:** 2025-09-01 | **Team:** Workspace Platform
- **Deliverables:** [\[UXR Report\] Unified Roster Foundational Interview with Admins \(go/ur-admin-uxr\)](#)

**Summary & Research Context:** This research report investigates the administrative challenges and opportunities for Google Workspace Group Administrators regarding the "Unified Roster" concept. Findings reveal that while a unified roster provides monumental value for simplifying end-user onboarding and access management, enterprise adoption is fundamentally blocked by

the pervasive lack of centralized visibility into group usage across connected services. Key roadmap recommendations include developing a centralized admin visibility cockpit, implementing simulation ("dry run") capabilities before group deletion, supporting mutable non-email shell groups, and enforcing granular Role-Based Access Control (RBAC) and bidirectional directory synchronization.

#### **Key Insights & Takeaways:**

- Administrators face a critical lack of centralized visibility into how groups are used across the organization, leading to over-provisioned access and zombie groups.
  - Key product opportunities include a Centralized Visibility Cockpit within the Admin Console and dry-run impact simulations before group deletion.
  - Provide mutable, non-email-addressable shell groups purely for access control and security management.
  - Enterprise adoption requires robust Organization Unit (OU)-based Role-Based Access Control (RBAC) and bidirectional directory sync capabilities.
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#### *4. IT Admins IAM Prioritization Survey (MaxDiff & TURF)*

- **Date / Status:** 2025-10-30 | **Team:** Workspace Platform
- **Deliverables:** [Quantifying IdP IAM Features UXR Report \(go/IAM-maxdiff\)](#)

**Summary & Research Context:** Privileged admin just-in-time (JIT) access is the top prioritized feature for IT decision-makers when choosing their IAM solution, followed by robust directory synchronization features like Password Hash Synchronization (PHS), and comprehensive MFA Configuration, significantly outweighing others like session duration policy or single logout. The primary feature gaps currently limiting GWS IdP adoption relate to lacking JIT access, on-premises interoperability, and advanced conditional access features. Most organizations favor a best-in-breed vendor strategy and face significant challenges managing employee changes, which currently outpaces funded roadmap capabilities for 2026.

#### **Key Insights & Takeaways:**

- MFA methods beyond SMS are highly valued, whereas GWS lacks location-based 2-step verification (trusted IPs) and device-based filters for authentication.
  - GCDS/GSPS is less reliable than Entra ID's PHS, leading to continued reliance on on-prem Active Directory (AD) as Source of Truth (SoT). GWS lacks bi-directional sync.
  - Privileged Admin Just-in-Time (JIT) Access (Provides temporary, time-bound admin privileges via an approval workflow) is rated as the single most important capability overall. JIT is enforced by 88% of admins, often via approval workflows.
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#### 5. *Workspace Identity: IAM Admin Needs Deep Dive*

- **Date / Status:** 2025-08-01 | **Team:** Workspace Platform
- **Deliverables:** [Contain MSFT Entra ID Compete IAM UXR Report \(go/compete-entra-uxr\)](#)

**Summary & Research Context:** Organizations prioritize Microsoft Entra ID because of its seamless integration with Windows ecosystems, bundled pricing models, and comprehensive security features like conditional access. Significant barriers to Google Workspace Identity adoption include perceived feature gaps in device management and administrative delegation, as well as the high financial and operational costs associated with migrating away from established identity providers.

#### **Key Insights & Takeaways:**

- Entra ID offers robust features, especially in Conditional Access (CA) combined with Multi-Factor Authentication (MFA) and seamless on-prem synchronization. In contrast, GWS lacks parity in table-stakes features and requires significant investment to compete.
  - Entra ID is often perceived as "free" due to bundling with Microsoft 365 E3/E5 licenses. High exit costs and investments in existing integrations financially prohibit switching, even if alternatives offer perceived benefits. GWS suffers from a perception of double-billing when used alongside Entra ID/M365, leading to a perceived increased Total Cost of Ownership (TCO). Cost optimization is a primary motivator for re-evaluation.
  - The primary factor is deep entrenchment in the Microsoft ecosystem (e.g., M365, Windows OS, on-prem Active Directory [AD]). Entra ID is seen as a "natural evolution" from on-prem AD.
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#### 6. *Guest Pass Admin Alpha Launch Interview*

- **Date / Status:** N/A | **Team:** Workspace

**Summary & Research Context:** No summary recorded.

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#### 7. *Guest Pass: Host Needs & Understanding Guest Labels*

- **Date / Status:** 2025-06-01 | **Team:** Workspace Platform
- **Deliverables:** [Deliverable](#)

**Summary & Research Context:** This report focuses on understanding host needs and interpretations of guest labels for external collaboration in Google Workspace.

#### **Key Insights & Takeaways:**

- We assumed users had sufficient clarity into what they had shared externally, but now we see that limited visibility into "who has access to what and at what level" creates a lack of confidence and control.
  - We previously expected users to manage access per doc, but now we believe a single, in-app page showing all permissions would streamline auditing and reduce errors.
  - Currently, controls are distributed among Drive apps and admin console. We recognize, through this research, that admins need a centralized, in-app hub to audit and adjust access levels more effectively.
  - Currently, individual document-level controls are designed to give admins and end users enough precision, but now we believe that manual checking and adjusting per doc access creates bottlenecks and inefficiency.
  - Icons paired with colors for labels aid understanding by adding more context to the colors, which can be interpreted differently.
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#### *8. Unified Roster Admin Exploration Design Workshop*

- **Date / Status:** N/A | **Team:** Workspace

**Summary & Research Context:** No summary recorded.

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#### *9. Unified Roster Admin Exploration*

- **Date / Status:** N/A | **Team:** Workspace

**Summary & Research Context:** No summary recorded.

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#### *10. Blind Research on Identity Gap with Prospective and Churning customers*

- **Date / Status:** N/A | **Team:** Workspace

**Summary & Research Context:** No summary recorded.

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#### *11. Agent AI Identity User Research*

- **Date / Status:** N/A | **Team:** Workspace

**Summary & Research Context:** No summary recorded.

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## 12. How Enterprise Companies Use Gemini

- **Date / Status:** 2025-01-29 | **Team:** Workspace Platform
- **Deliverables:** [How Enterprise Companies Use Gemini for Work](#)

**Summary & Research Context:** Experts have a strong attachment to Gemini, which is transforming their work and yielding productivity gains, especially with standalone tools. Diverse and unique use cases highlight Gemini's adaptability, but the quality of in-context tools within Workspace apps is behind standalone versions. Limitations in image generation, presentation drafting, and document formatting prevent users from fully utilizing Gemini's potential.

### Key Insights & Takeaways:

- Users crave enhanced collaboration and knowledge sharing features within Gemini, indicating a gap between its current capabilities and the collaborative nature of enterprise work. It is known that collaboration is key in enterprises. This insight highlights a specific unmet need within the Gemini ecosystem for better sharing of AI-generated assets and workflows, which is currently a point of friction. Individual productivity was the main Gemini value, but now we see a strong demand for it to become a collaborative platform.
  - Standalone Gemini applications (like Gemini.google.com, Gems, NotebookLM) are unlocking transformative productivity gains by enabling complex, cross-functional tasks that were previously impractical, despite a quality lag in in-context versions. It was understood that different interfaces might have different strengths, but this insight pinpoints that standalone tools are not just preferred for certain tasks, but enable previously impossible workflows, indicating a step-change in productivity rather than just incremental improvement.
  - Gemini's integration into work contexts fosters deep user attachment, making it an indispensable tool rather than just a feature. At the thought of losing access, some users expressed sadness or anxiety, suggesting a deeper integration than typical software adoption. We might have thought Gemini is a useful add-on, but now we see it's becoming fundamental to users' workflows because Gemini saves significant time, transforms work processes, fosters collaboration, and ability to understand work context.
  - The vast majority of Gemini use cases observed in enterprises are unique, demonstrating its broad adaptability, yet current limitations in image generation, presentation drafting, document formatting, and side panel functionality hinder its full potential. While AI tools are expected to be versatile, the finding that 77 out of 80 identified use cases were unique is a strong indicator of Gemini's exceptional adaptability to diverse, specific work needs. Gemini's application at work is highly personalized and widespread across varied functions.
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### 13. Investigating into Admin needs for guest account managements

- **Date / Status:** 2024-05-31 | **Team:** Workspace
- **Deliverables:** [Visitor Account Admin Needs UXR \(go/visitor-account-uxr\)](#)

**Summary & Research Context:** This research tapped into how SMB, MM and Enterprise admins manage external guests when the counterparts do not share the same productivity Suites. It revealed customers' willingness to pay for managing external guests, and challenges of managing external guests because the efforts onboarding and auditing are most strenuous. What could differentiate GWS guest solution is to automate cross-org provisioning with inbound SCIM, delegate guest account management to internal sponsors, and highlight our compliance in Public Sectors.

#### Key Insights & Takeaways:

- External collaboration is so prevalent and critical that customers are willing to pay for it and are looking for 3P solutions.
  - What could differentiate GWS guest solution is to automate cross-org provisioning with inbound SCIM, delegate guest account management to internal sponsors, and highlight our compliance in Public Sectors.
  - Most important admin CUJs for external collaboration are onboarding, auditing and offboarding external accounts in AC, where onboarding and auditing are most strenuous and challenging.
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### 14. AuthZ Role Management Exploratory Research

- **Date / Status:** 2024-10-31 | **Team:** Workspace
- **Deliverables:** [AuthZ Role Management Exploratory UXR \(go/authz-uxr\)](#)

**Summary & Research Context:** SMB and Enterprise customers employ different strategies for admin management controls, especially with Super Admin role assignment. Our findings revealed security risks of over-granting permissions due to the frictions in creating new admin roles and management role changes, despite that all admins were aware of the importance of delegating admin roles with least privileges.

#### Key Insights & Takeaways:

- Super Admins go through 4 user journeys of assigning, creating, changing and auditing admin roles, among which role creation is most challenging, whereas role assignment is most error prone and precautions.
- Role creation is hard & time-consuming because 1) Privilege descriptions are unclear and hard to comprehend. 2) Creating a new role takes rounds of testing and validations.



- The Paradox of Super Admins revealed the problems of having too many and too few Super Admins, and revealed limitations in GWS Role Assignments, such as (1) certain privileges are baked in Super Admins (2) Lack of assignment scope to OU lead to over-granting permissions.
  - SMB SAs leverage system roles as much as they can, whereas enterprise SAs prefer to create and assign custom roles.
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## **Theme 2: Frontier Generative AI, Multimodal Creation & Visual Tools (Pics & Editors)**

**Focus:** Generative AI content creation, Multimodal creation (ImageGen, Pics, Video editing, On-canvas direct manipulation), AI onboarding for low-AI-literacy creators, Competitive intelligence (CapCut), and AI & Memory.

**Total Studies:** 12

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### *1. AI Opportunities with Visual Content Generation Study for Low AI Literacy Users*

- **Date / Status:** 2026-08-03 | **Team:** Workspace
- **Deliverables:** [AI Opportunities with Visual Content Generation Study](#)

**Summary & Research Context:** This qualitative study investigates how full-time professionals create and edit visual content, identifying workflow friction points and barriers to adopting AI visual generation tools. By analyzing end-to-end visual workflows, the research highlights key areas where current AI image tools fall short and outlines actionable product opportunities to better support user needs.

### **Key Insights & Takeaways:**

- Bridging the design expertise gap Though participants found Canva and AI tools user-friendly, they often struggled to create polished, cohesive designs. Current tools offered limited guidance, while AI image generation required too many iterations and provided limited control over targeted changes.
- Managing fragmented feedback Collaboration between stakeholders and visual creators brought together the expertise and approvals needed for higher-stakes work, but feedback was often delayed, conflicting, or scattered across channels. This created misalignment and slowed the design process.
- Articulating the vision Respondents without design backgrounds often needed visual examples to discover and communicate what they wanted, while creators struggled to interpret vague or evolving requests. Mood boards and inspiration images helped bridge this gap but took considerable time.

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## 2. Visual Creator Journeys and Needs Among Users with Low AI Literacy

- **Date / Status:** 2026-08-03 | **Team:** Workspace
- **Deliverables:** [AI Opportunities with Visual Content Generation Study](#)

**Summary & Research Context:** No summary recorded.

### Key Insights & Takeaways:

- Managing fragmented feedback Collaboration between stakeholders and visual creators brought together the expertise and approvals needed for higher-stakes work, but feedback was often delayed, conflicting, or scattered across channels. This created misalignment and slowed the design process.
- Bridging the design expertise gap Though participants found Canva and AI tools user-friendly, they often struggled to create polished, cohesive designs. Current tools offered limited guidance, while AI image generation required too many iterations and provided limited control over targeted changes.
- Articulating the vision Respondents without design backgrounds often needed visual examples to discover and communicate what they wanted, while creators struggled to interpret vague or evolving requests. Mood boards and inspiration images helped bridge this gap but took considerable time.

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## 3. Pics HaTS Topline Research Findings

- **Date / Status:** 2026-08-07 | **Team:** Editors
- **Deliverables:** [Pics Standalone HaTS topline report \(under review\)](#)

**Summary & Research Context:** Analysis of the recent Pics HaTS shows a clear satisfaction gap between consumer and business users. Consumer respondents report significantly higher overall satisfaction than business users, with an 84% CSAT (12% DSAT) compared to 65% for business users (23% DSAT). By mapping the 5-point satisfaction levels to a 1-5 scale, mean satisfaction scores are 4.23 out of 5.0 for consumers and 3.77 for business users (+0.47 difference on the mean satisfaction scores)\*.

### Key Insights & Takeaways:

- The gap between consumers and business users' satisfaction lies in their workflow complexity and tolerance of precise controls: Business users attempt precise image edits, product mockups, and client deliverables, where auto-segmentation errors and quality loss create friction.

- Prosumer usage in consumer data: A notable portion of self-identified "consumer" respondents actually use Pics for commercial tasks like real estate photography, film production design, engineering diagrams, and marketing collateral.
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#### 4. *Pics Dogfood CSAT Survey*

- **Date / Status:** 2026-07-18 | **Team:** Workspace Platform
- **Deliverables:** [Pics Dogfood CSAT Survey Feedback \(Under Review\)](#)

**Summary & Research Context:** We analyzed dogfood feedback from 308 valid respondents in our recent CSAT survey (launched June 25 – July 9, 2026) to evaluate product readiness for standalone Google Pics prior to GA. The research investigated satisfaction drivers, and usage patterns, and revealed feature gaps, and how Pics fits into users' existing workflows alongside alternative visual creation tools.

#### **Key Insights & Takeaways:**

- Pics achieved an 83% CSAT score (and impressively low DSAT 8%) meeting the GA target, with users praising its ease of use, automatic object detection, and localized text and image refinement.
  - When asked about Dogfood users' alternative visual creation tools, Pics is currently an additive tool rather than a replacement for other tools. Currently, 79% of users still retain their legacy tools (e.g., Gemini, Photoshop, Google Slides) due to missing table-stakes features (cropping, resizing, transparent PNGs) and a lack of deep ecosystem integration. Personal and casual use (63.4% photo editing/personal projects, 44.7% for fun) reflects early low-stakes experimentation, whereas satisfaction levels related to business related use cases score lower.
  - Dissatisfaction and workflow friction stem primarily from AI model issues—such as quality decay across iterative edits, spatial inaccuracy, and hallucinations—alongside missing basic editing utilities like cropping, resizing, and transparent PNG exports.
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#### 5. *Pics Dogfood CSAT survey results 2026/07*

- **Date / Status:** 2026-07-26 | **Team:** Editors
- **Deliverables:** [Pics Dogfood CSAT Survey Feedback \(Under Review\)](#)

**Summary & Research Context:** No summary recorded.

#### **Key Insights & Takeaways:**

- Despite high satisfaction, Pics is currently an additive tool rather than a replacement for other tools. Dogfood users are forced back to other tools due to missing fundamental utilities (cropping, resizing, transparent PNGs).
  - Dogfood users struggle when results are unusable due to model issues, lack of basic editing features, or quality decay across iterations. When image outputs feel unpredictable, inaccurate, or change areas outside the user's selection, frustration grows.
  - High CSAT (83%), Low DSAT (8%) Core UI and localized editing mechanics are highly intuitive. The fundamental interaction model with object segments and localized edits is validated and beloved by Dogfooders for rapid iterations.
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#### 6. Log Insights for Pix: ImageGen 3P in Workspace Usage

- **Date / Status:** 2026-03-28 | **Team:** Workspace
- **Deliverables:** [Deliverable](#)

**Summary & Research Context:** Log data analysis reveals that Google Workspace users frequently integrate third-party visual design apps like draw.io, Figma, and Miro to fulfill creative needs. Despite this demand, a lack of full integration from many image generation players often forces users to leave the Workspace environment to complete their visual tasks.

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#### 7. Visual Creation Results from NYC AI Onboarding Immersion Onsite

- **Date / Status:** 2024-04-08 | **Team:** Editors
- **Deliverables:** [Deliverable](#)

**Summary & Research Context:** Small businesses in New York City utilize generative AI for visual creation but face significant friction from branding inaccuracies and disjointed mobile workflows. Successful adoption depends on shifting from pure image generation to intelligent extraction and precise reference grounding to ensure brand consistency.

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#### 8. AI Usage in Nonprofit Marketing and Visual Content Creation

- **Date / Status:** 2026-03-27 | **Team:** Editors
- **Deliverables:** [Deliverable](#)

**Summary & Research Context:** Nonprofits primarily utilize AI for marketing and content creation to improve speed and efficiency, with Canva being the most dominant tool across all regions. The majority of organizations prefer using AI features embedded within their existing design workflows rather than standalone generative tools. Adoption is driven by the desire for

productivity gains, creative freedom, and cost reduction, though human representation remains preferred for beneficiary imagery.

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#### 9. CapCut Video Editor Competitive Analysis Deep Dive

- **Date / Status:** 2026-05-15 | **Team:** Editors
- **Deliverables:** [CapCut Video Editor Teardown and Desk Research \(Under Review\)](#)

**Summary & Research Context:** CapCut dominates the social media video editing market by providing an all-in-one ecosystem of AI-powered tools, templates, and assets that drastically reduce production time for creators. Professional users demonstrate a high willingness to pay for premium features like auto-captions and 4K exports, while aggressive monetization of previously free utilities has created a trust crisis among casual hobbyists. Integrating native editing capabilities into Google Drive could prevent the significant user churn caused by the current 'download-edit-reupload' cycle required to use third-party tools.

#### Key Insights & Takeaways:

- Locking the highly popular auto-caption feature behind a premium paywall has become a leading driver of creator outrage and community backlash.
  - Professional business creators easily absorb \$20/mo subscriptions as they calculate value by time saved, whereas casual and hobbyist users are price-sensitive.
  - CapCut wins the market with Comprehensive Natively Built-in Tools: It provides an all-in-one package for video editing, including template, auto-caption, and AI features, even for image editing
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#### 10. ImageGen Research Exchange: Literature Review & XFN Intake

- **Date / Status:** 2026-02-27 | **Team:** Editors
- **Deliverables:** [Deliverable](#)

**Summary & Research Context:** This hub synthesizes external literature and internal insights on image generation design and user trends for the Pix team. It highlights how users perceive AI latency as a sign of creative effort and explores the efficiency of direct manipulation interfaces over standard conversational prompting.

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#### 11. On-Canvas Editing Interview Study

- **Date / Status:** 2026-04-01 | **Team:** Editors
- **Deliverables:** [\[Pix UXR\] On-Canvas Editing Interview Study 2026](#)

**Summary & Research Context:** This study evaluates the discoverability and intuitiveness of on-canvas generative AI image editing controls within Pix, highlighting a strong user preference for direct GUI manipulations over natural language prompts. Findings reveal significant friction points, including ambiguous system states, an inefficient 'Apply' bottleneck for single edits, and 'collateral damage' where localized changes unpredictably alter unselected image areas.

**Key Insights & Takeaways:**

- The "Apply" Bottleneck → All users tripped over this two-step workflow for batch editing paradigm. Users inherently expect immediate execution and real-time feedback for single local edits.
  - Ambiguous System States & Poor discoverability → White dots, “detecting objects text” etc type of SIs are confusing.
  - Unintended “Collateral Damage” & Lack of Localized Precision → Users strongly gravitate towards UI direct manipulations over prompts to convey desired changes, and appreciate local edits as specified.
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*12. Position Paper: AI and Memory*

- **Date / Status:** 2025-08-15 | **Team:** Workspace
- **Deliverables:** [Diamond Position Paper: AI + Memory](#) 🧠 ([go/ai-memory-paper](#))

**Summary & Research Context:** This position paper synthesizes insights from across Workspace and Google on the topic of AI and memory in the work context.

From these insights, this paper provides 4 actionable recommendations for teams building memory and personalization related features in Workspace:

- 1) Memory is a big gap for Gemini in Workspace.
  - 2) Users want WS Gemini to remember 3 types of memories: informational context, style preferences, and behavioral patterns.
  - 3) Shared memory has benefits, but also brings challenges.
  - 4) Control over memory and transparency into how it's used are important.
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**Theme 3: Multi-Sided Platform Extensibility, APIs & App Ecosystems**

**Focus:** 3P & 1P Developer ecosystems, Chat apps, Workflows, Workspace APIs, Marketplaces, and Superapp concepts across Admins, Developers, and End-Users.

**Total Studies:** 17

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### 1. Data Marketplace CSAT

- **Date / Status:** 2021-03-01 | **Team:** Data
- **Research Archive Link:** [Data Marketplace CSAT](#)
- **Deliverables:** [walkthrough slides/deck](#), [Data Marketplace CSAT 2020 Detailed full report, summary](#)

**Summary & Research Context:** The goal of this survey was to measure user satisfaction and pain points, tracking product excellence metrics and informing OKR priorities. Active users were sent a link for a three-question survey containing a Likert scale and two open-ended questions. Of the 1314 users who received the survey, 142 surveys were completed (n=142) between Sep. 29-Oct. 14, 2020 (10.8% response rate). Of the completed surveys, 84 respondents provided open-ended responses. A marginal change from 2019 (M=86.1, SD=18.6) to 2020 (M=80.2, SD=23.7) was observed based on a Welch T-test ( $t(206)=-1.96$ ,  $p = 0.05$ ). Three user frictions were identified.

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### 2. Superapp Concepts Evaluation

- **Date / Status:** 2023-09-01 | **Team:** Workspace
- **Research Archive Link:** [Superapp Concepts Evaluation](#)
- **Deliverables:** [Deliverable](#)

**Summary & Research Context:** This study aimed to evaluate the discoverability, usability, and desirability of three post-MVP ideas for Super App [On-Demand Translation, Polling, Scheduled Send]

#### Key Insights & Takeaways:

- The three Superapp concepts evaluated were perceived as native, unobstructive experiences in Chat
  - Participants felt the addition of Translation, Polling, and Schedule Send features were discoverable and accessible and not obstructive to their every day tasks
  - On-Demand Translation was viewed as the least applicable tool, as most participants did not frequently communicate with people who did not speak the same language “I think it's a useful feature. I mean it's not a ‘My God, where's this been my whole life?’ type feature.” -P1 (Re: On-Demand Translation)
  - Participants were excited about using Schedule Send to communicate across time zones, and liked the idea of creating Polls to make decisions in conversations
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### 3. Chat Auto Workflow

- **Date / Status:** 2023-08-01 | **Team:** Workspace
- **Research Archive Link:** [Chat Auto Workflow](#)
- **Deliverables:** [Deliverable](#)

**Summary & Research Context:** This research aimed to uncover user understanding of Workflow Builders and evaluate the usability of three Workflow Builder options

#### Key Insights & Takeaways:

- An initial walkthrough or demo of the feature would help with understandability and discoverability
  - Participants expected the automations would help alert them to activity occurring across Workspace tools (eg. Drive, Calendar)
  - The Inline mode provided participants with the most sequential guidance and was the most preferred option
  - Chat is often used and kept open, so participants felt that notifications, reminders, and alerts in Chat through an automation 'Bot' would be highly effective
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### 4. 3P Workspace Developer Research Summit Talk

- **Date / Status:** 2024-02-01 | **Team:** Gmail-Hub
- **Research Archive Link:** [3P Workspace Developer Research Summit Talk](#)
- **Deliverables:** [Why Workspace Developers Build? A Deep Dive into Chat App Developers' World](#)

**Summary & Research Context:** To incentivize Workspace developers, we need to identify expedited CUJs for each Workspace products based on use case fit and showcase exponential growth opportunities for different contexts. Enabling developers to unlock business opportunities fuels more organic integration efforts. We should also make the areas of customer demand, reusable blocks & partnership gain more explicit.

#### Key Insights & Takeaways:

- Partners build chat apps for expedited CUJs & distribution channels.
  - Enabling developers to unlock business opportunities fuels more organic integration efforts.
  - Public app developers build for customer demand, reusable blocks & partnership gain.
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#### 5. Chat apps: Discoverability, usability, and friction points in the user journey

- **Date / Status:** 2023-12-01 | **Team:** Workspace
- **Research Archive Link:** [Chat apps: Discoverability, usability, and friction points in the user journey](#)
- **Deliverables:** [Deliverable](#), [Deliverable](#)

**Summary & Research Context:** In three semi-structured observational tasks, 8 Chat users found, installed, and used chat apps, with the goal of identifying friction points in the Chat app user journey.

A few key findings:

- Users exerted more effort than expected finding any of the several Chat app entry points. The most common method to discover entry points was to exit Chat and use Google Search, revealing a need for help in knowing where to start looking. But once a single entry point was found, it was reused for all later tasks, making the whole Chat app journey feel relatively easy by study's end.
- Users disproportionately depended on the search features of Marketplace and Bot Catalog to find apps; 21 out of 24 journeys involved search in some way. Users rarely searched for apps by name but searched by keywords related to the problem they were trying to solve, which did not yield helpful app suggestions.
- Users were intimidated by the process of invoking Chat apps, because they lacked familiarity with using syntax-based interfaces. Welcome statements from Chat apps were useful starting points, but they did not help with troubleshooting mistakes, which made the process of testing apps slower and more frustrating than desired.

#### Key Insights & Takeaways:

- The most common method for discovering Bot Catalog and GWM was to exit Chat and use Google Search, suggesting a need for improved orientation within Chat itself.
- Users disproportionately depended on search (vs. browsing) when choosing apps, but rarely searched for apps by name and in general found the search feature to be less helpful than expected.
- A lack of familiarity with text-based commands made the process of interacting with chat apps intimidating and at times frustrating.

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#### 6. Workflow Builder usability and comprehension test

- **Date / Status:** 2023-11-01 | **Team:** Workspace
- **Research Archive Link:** [Workflow Builder usability and comprehension test](#)

- **Deliverables:** [Deliverable](#), [Deliverable](#)

**Summary & Research Context:** The primary objectives of the research were to assess comprehension and usability of the current iteration of Workflow Builder (as of Oct/Nov, 2023). Participants provided evaluations of Workflow Builder's interface, menus, and templates, and were asked to build a workflow automation (standardized across all participants).

**Key Insights & Takeaways:**

- The Workflow Builder interface was easy to learn, but participants were challenged by a lack of understanding of core automation concepts such as triggers, steps, and sequences. This was true across all levels of prior experience using workflow automation tools, although those without prior knowledge were slower to comprehend.
  - Workflow Builder templates generated more excitement about automation than discussing automation in the abstract, suggesting that future user engagement will depend on the quality of the templates available immediately.
  - Being a Google product enhanced usability because of its familiarity to other Google apps, but it drew unexpected attention to a lack of first-party and third-party integrations, which were strongly desired by participants.
- 

*7. Perceptions of Super App when it communicates via DM*

- **Date / Status:** 2023-10-01 | **Team:** Workspace
- **Research Archive Link:** [Perceptions of Super App when it communicates via DM](#)
- **Deliverables:** [Deliverable](#), [Deliverable](#)

**Summary & Research Context:** This study aimed to understand users' perceptions of Super App in five different use cases in which Super App communicates as itself, via DM. These included sending a welcome message, sending reminders, managing requests to join, notifying of new features, and sending workflow notifications. Along with evaluations of these specific use cases, participants provided overall assessments of Super App: what they expect it to do, how they expect it to interact, and why they think it is there.

**Key Insights & Takeaways:**

- Super App sending feature updates via DM was seen as helpful. But these DMs had the most potential to be perceived as spam-like, and participants desired control over the frequency and content of these updates.
- Reminder setting was received favorably but reminder delivery was not flexible and dynamic enough for test participants' needs.

- Super App was seen overall as helpful but combining low-priority messages (e.g., feature updates) with high-priority messages (e.g., reminders) was seen as disruptive and made users question whether new messages from Super App would be worth checking.
  - Super App representing itself as “Chat App” in the DM roster was incomprehensible to test participants, but Chat App’s welcome message provided sufficient context and information once discovered.
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#### 8. Understanding Chat App Developers' Motivations and incentives

- **Date / Status:** 2023-06-01 | **Team:** Workspace
- **Research Archive Link:** [Understanding Chat App Developers' Motivations and incentives](#)
- **Deliverables:** [Chat App Builders' Incentives Interview \(biz perspectives\)](#)  
[\(go/dev-incentives-uxr\)](#)

**Summary & Research Context:** Following up on the quantitative survey, this interview study (N=19) uncovered why developers build on chat platforms after all, and what comes to mind when they decide on building on platform over another, including both business incentives and technical costs.

#### Key Insights & Takeaways:

- 🌟 - Partners build chat apps for expedited CUJs, distribution channels and low overheads, esp in the verticals of ITSM, HR and Collaborative Creation. 📈 - Public app developers build for customer demand, user reachability & partnership gain, the former two of which work to our disadvantage. 😊 - Google Chat appeals to developers & partners for its growth potential advantages as part of holistic Workspace Suite & partnership value. 👍 - Partners expect to see platform feature parity, esp wrt exposing app discovery in more surfaces, reducing hurdles in publishing apps, and reusing components as part of GWS.
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#### 9. Marketplace UXR: End users, Admins and Developers' needs

- **Date / Status:** 2023-06-01 | **Team:** Gmail-Hub
- **Research Archive Link:** [Marketplace UXR: End users, Admins and Developers' needs](#)
- **Deliverables:** [Marketplace UXR: End users, Admins and Developers' needs](#)

**Summary & Research Context:** End users found friction app discovery & installation because of the difficulty in locating app store / marketplace pop-up, lack of global search surfaces for 3P apps, and the lack of searching capabilities in marketplace; Admins found search capabilities the topmost painpoint, followed by concerns with app quality and upfront charge potentials;

developers expected to have more featured app section and more promotion venues with shared customers.

### **Key Insights & Takeaways:**

- Developers expected to have more featured app section and more promotion venues with shared customers.
- Admins found search capabilities the topmost painpoint, followed by concerns with app quality and upfront charge potentials;
- End users found friction app discovery & installation because of the difficulty in locating app store / marketplace pop-up, lack of global search surfaces for 3P apps, and the lack of searching capabilities in marketplace;

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#### *10. Google Chat Customers' App Integration Needs Survey*

- **Date / Status:** 2023-10-01 | **Team:** Gmail-Hub
- **Research Archive Link:** [Google Chat Customers' App Integration Needs Survey](#)
- **Deliverables:** [Top Enterprise integration needs for Google Chat Survey Report \(go/top-integration-survey\) 2023.10.10](#)

**Summary & Research Context:** The survey revealed Chat customers' expectations on both internal Workspace and external app integrations: For 1P integration points, Gmail, Google Meet and Google Docs are most needed within Google Chat; For 3P integration points, ITSM (ServiceNow), Task management (Workast, Monday.com), File management system (OneDrive, DropBox) and Human capital management (SAP Successfactors, Workday) are among the apps showing the highest helpfulness potential with a Google Chat integration. In addition, prospective Chat customers' reported missing integrations aligned with the external app needs (e.g. ServiceNow, Dropbox). The insights help us shape our Business strategy for the 2024 planning

### **Key Insights & Takeaways:**

- Qualitative data supports the focused areas on Task management, video conferencing apps, CRM and HCM vertical. Google Chat users also express interest in Gen AI tools & notification feeds for social media & news, and interop support from other chat alternatives as potential integrations for Google Chat.
- GWS products (esp. Gmail, Google Docs & Meet), show high potential to be helpful as an integration in Google Chat, while also reaching the largest group of current Chat users. Google Meet is perceived with highest average helpfulness scores and aligns with usage trends seen for the Zoom integration in Google Chat.
- Among Google Chat users, ITSM1 (ServiceNow), Task management (Workast, Monday.com), File management system (OneDrive, DropBox) and Human capital

management (SAP Successfactors, Workday) are among the apps showing the highest helpfulness potential with a Google Chat integration.

- 65% of non-Google Chat users reported that having additional integrations available for their common work apps would influence their decision to use Google Chat. Human capital management apps (SAP SuccessFactors, Workday) show above average influence, (similar to usefulness trends seen for Google Chat users) and Workday, followed by various types of apps, such as Dropbox, ServiceNow and Hubspot.

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#### 11. Drivebot CSAT & Kano Model-based Feature Prioritization

- **Date / Status:** 2023-02-01 | **Team:** Gmail-Hub
- **Research Archive Link:** [Drivebot CSAT & Kano Model-based Feature Prioritization](#)
- **Deliverables:** [Drivebot Satisfaction and Feature Prioritization](#)

**Summary & Research Context:** :drive: Drivebot CSAT and Feature Prioritization Survey :chat: To ensure excellent 1P & 3P app integration in Chat, we conducted the first-ever Chat App End User Satisfaction survey study for Drivebot, one of the most used chat apps in Google Chat, in addition to prioritizing features based on the Kano model. The study validated proposed features for H1 improvements and revealed new focus areas, such as notification management and organization.

:love-1: Drivebot end users are very satisfied (top box = 91.5%), regardless of their usage frequency. 🙌 - Drivebot is loved for its instant notification, increased efficiency and assured attention span & history tracking in Chat. 😊 - Drivebot in Chat assures users of receiving timely notification, while lacking support of selective information consumption and notification management. 😞 - Lack of notification control, proper organization and sensemaking support may lead to complete ignorance and jeopardize users' productivity. Take a look at all insights and recommendations for improving Drivebot at [go/drivebot-csat](#)

#### Key Insights & Takeaways:

- Users love Drivebot (91.5% top box satisfaction), but may ignore it if the notifications are spammy and too hard to manage.
- Lack of notification control, proper organization and sensemaking support may jeopardize users' productivity and discourage interaction (101/340).

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#### 12. Chat Automated Workflow Use Case Explorations

- **Date / Status:** 2023-07-01 | **Team:** Gmail-Hub
- **Research Archive Link:** [Chat Automated Workflow Use Case Explorations](#)
- **Deliverables:** [Deliverable](#)

**Summary & Research Context:** To inform product strategy and priorities with building MVPs of Chat Automation Workflow builder, we interviewed n=14 chat end users to understand why and how they build and use workflow automations that relate to chat platforms. 🌟 - Chat-related workflows primarily take place around message (e.g. reply reminder, keyword, or timestamp), rarely triggered from emoji reaction. 🧑 - Workflow builders may or may not be workflow users; however, builders and users coordinate to make the automation work and often co-own a workflow. 👁️ - Most builders (or users before conversion into builders) struggle most with getting started and need assistance (e.g. templates, in-process prompts). 🏆 - Most frequent co-used products in chat automated workflows are Form submission, Drive storage, and Gmail.

### Key Insights & Takeaways:

- 👁️ - Most builders (or users before conversion into builders) struggle most with getting started and need assistance (e.g. templates, in-process prompts).
  - 🏆 - Most frequent co-used products in chat automated workflows are Form submission, Drive storage, and Gmail.
  - 🌟 - Chat-related workflows primarily take place around message (e.g. reply reminder, keyword, or timestamp), rarely triggered from emoji reaction.
  - 🧑 - Workflow builders may or may not be workflow users; however, builders and users coordinate to make the automation work and often co-own a workflow.
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### 13. How Googlers Develop within Chat: G3 Assist Exploratory Research

- **Date / Status:** 2022-10-01 | **Team:** Gmail-Hub
- **Research Archive Link:** [How Googlers Develop within Chat: G3 Assist Exploratory Research](#)
- **Deliverables:** [How Googlers Develop within Chat: G3 Assist Exploratory Research \(go/g3a-cujs\)](#), [G3 Assist UX Research Proposal \(Stage I: Exploratory\)](#)

**Summary & Research Context:** Based on 60 minutes in-depth interview with Googlers from diverse PAs (n = 10), the study uncovered 5 end user critical journeys (CUJs) that illustrate users' goals of installing (or onboarding into a chat app), collaborating, consuming unfurled content without clicks and context switching, immediate action-taking and notification management.

### Takeaways:

- Spammy welcome messages, redundant installation , and friction in finding where to install prevent users from adoption
- Silent failures, unclear expiration due date and debugging friction stop users from making the best of G3A chat app

- More salient interactive affordances are needed for users to take immediate actions (e.g. 3 out of 10 users noticed DM features, 7 never used editing features partly due to feature discovery).

#### Recommendations:

- Offer failure warning and actionable debugging guide (stop silent failures)\*
- Allow for silent installation or domain installation to prevent spamming users
- Onboard users with more salient affordances (e.g. DM with G3A, 1:1 with G3A)
- Expand G3A coverage for more links and with more interactivity
- Redesign UI kit with more compact cards

#### Key Insights & Takeaways:

- Richer contextual information and more salient affordances inform immediate actions if needed.
- Spammy welcome messages, redundant installation, and friction in finding where to install prevent users from adoption
- Silent failures, unclear expiration due date and debugging friction stop users from making the best of G3A chat app

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#### 14. Data Marketplace Exploratory Study 2019 Q4

- **Date / Status:** 2020-03-01 | **Team:** Data
- **Research Archive Link:** [Data Marketplace Exploratory Study 2019 Q4](#)
- **Deliverables:** [Deliverable](#), [Deliverable](#), [Deliverable](#)

**Summary & Research Context:** To explore Data Marketplace Explorer (DME) use cases and friction points to provide scoping for further studies, we conducted 60-minute artifact interviews over GVC with 9 participants to address two research questions: 1). How do users discover and choose a new data source?; 2). What information do users need to make decisions about integrating with an existing data source. Based on thematic analysis, four takeaways by stage and six CUJs were generated.

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#### 15. Onboarding Chat Users with Extensible Landscape: Research with Chat Apps

- **Date / Status:** 2022-09-01 | **Team:** Gmail-Hub
- **Research Archive Link:** [Onboarding Chat Users with Extensible Landscape: Research with Chat Apps](#)
- **Deliverables:** [Onboarding Chat Users with Extensible Landscape: Research with Chat Apps](#)

**Summary & Research Context:** Google chat users have three stages in their chat integration onboarding journeys, and face various challenges through the process. The deck leveraged two empirical studies to reveal the onboarding framework for chat app end users across three platforms, and pointed out the frictions users face when onboarding, including locating the marketplace, search for apps, and also installation hurdles.

**Key Insights & Takeaways:**

- Chat app users go through vetting, inspection, installation and learning process when onboarding themselves into new chat apps.
  - Google Chat needs to better support users about the discovery and search journeys of the available chat apps by means of enabling domain installation, reducing the spammy messages from the installation confirmation and instructions across multiple contexts, and also better onboarding workflows that educate users about how to discover, find and install apps.
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*16. Chat App Developers' Adoption Drivers and Barriers Research - Part I, Survey*

- **Date / Status:** 2023-04-01 | **Team:** Gmail-Hub
- **Research Archive Link:** [Chat App Developers' Adoption Drivers and Barriers Research - Part I, Survey](#)
- **Deliverables:** [Chat App Developers' Adoption Drivers and Barriers Research - Part I, Survey](#)

**Summary & Research Context:** If Google Chat is going to become one of the best platforms for chat apps, we need to offer an excellent developer experience! We surveyed n=146 chat app developers from mixed backgrounds and recruitment channels to understand how they choose which chat platforms to build amongst Slack, Teams and Google Chat.

**Key Insights & Takeaways:**

- Upcoming (pre-GA) features have been long-awaited and will attract developers. E.g., the ability to initiate msg to end users from chat app and to read msgs without being mentioned will provide an inducement to build apps for Chat, so we should highlight these features in GA announcements.
- Public chat app developers determine where to build based on market value, whereas private app developers prioritize documentation & platform reliability, offering integration platforms different strategies to accommodate various development partners.
- Dissatisfaction with building apps for Google Chat is caused by feature parity issues, hard-to-use Docs (esp. DPP instructions), and constrained Card layout.



- Developers like Google Chat (esp. with the upcoming pre-GA Chat APIs)! Our CSAT is comparable with Slack (~73%) and 10% higher than Teams (but our DSAT is also +5%).
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#### 17. All about APIs V0 (Workspace API HaTS on Dev Docs)

- **Date / Status:** 2022-07-01 | **Team:** Gmail-Hub
- **Research Archive Link:** [All about APIs V0 \(Workspace API HaTS on Dev Docs\)](#)
- **Deliverables:** [All about APIs V0 \(2022Q2\)](#)

#### Summary & Research Context:

- Developers' satisfaction with Workspace APIs (five products APIs tested) were measured for the first time, with the Satisfied percentage ranging from 56.6% to 71.6%, which indicated room for improvements compared with Cloud API (87%) and other 3P integrations (see go/3p-devsat)
- Gmail API and Drive API have almost One Third of Dissatisfied users among respondents.
- API configuration is difficult for all Workspace products, especially for Authentication
- API is also hard to learn due to many specific areas for document improvements, such as lack of coding examples and difficulty to navigate

#### Key Insights & Takeaways:

- Developers' satisfaction with Workspace APIs (five products APIs tested) were measured for the first time, with the Satisfied percentage ranging from 56.6% to 71.6%, which indicated room for improvements compared with Cloud API (87%) and other 3P integrations
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#### Theme 4: Core Developer Infrastructure, Data Tools & Quantitative Benchmarking

**Focus:** Headless developer infrastructure, data storage engines, SQL tooling, indexing pipelines, and multi-year quantitative CSAT benchmarking.

**Total Studies:** 18

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#### 1. Evals CSAT 2020 Q4

- **Date / Status:** 2021-03-01 | **Team:** Data
- **Research Archive Link:** [Evals CSAT 2020 Q4](#)
- **Deliverables:** [report](#), [preso](#), [summary](#)

**Summary & Research Context:** The goal of this survey was to measure user satisfaction and pain points, tracking product excellence metrics and informing OKR priorities. Users who had run more than 5 Evals in the past 90 days were sent a link for a three-question survey containing a Likert scale and two open-ended questions. Of the 1841 users who received the survey, 656 surveys were completed (n=656) between Nov. 11-Nov. 30, 2020 (35.6% response rate). Of the completed surveys, 532 respondents provided open-ended responses. Users' perception about Documentation (-4.63%) and Configuration (a proxy for front-end improvements, -4.01%) have improved, whereas issues around Latency (+9.77%), Reliability (+4.75%), Interface (+9.80%) and Debugging (+8.07%) have persisted or worsened since the CSAT in Q2 2020. The interface-related complaints mainly stemmed from the EAS (52/532) and confusing navigation and searching experiences (12/532).

#### **Key Insights & Takeaways:**

- Users still face hurdles in deciding on appropriate setups (esp. when starting from scratch) in Evals (160/532).
- Users' perception about Documentation (-4.63%) and Configuration (a proxy for front-end improvements, -4.01%) have improved, whereas issues around Latency (+9.77%), Reliability (+4.75%), Interface (+9.80%) and Debugging (+8.07%) have persisted or worsened since the CSAT in Q2 2020. The interface-related complaints mainly stemmed from the EAS (52/532) and confusing navigation and searching experiences (12/532).
- High latency issues have caused productivity loss because of delays in starting Evals, running Evals and UI sluggishness (185/532). The delay in starting evals (71/532) involved a holdup in acquiring slots (35/532) and queuing congestion (30/532).
- Instability (187/532), noise (59/523), and high overheads of eventual failures continue reducing users' sense of control and productivity (230/532).

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#### *2. Rasta CSAT 2019 Q4*

- **Date / Status:** 2020-01-01 | **Team:** Data
- **Research Archive Link:** [Rasta CSAT 2019 Q4](#)
- **Deliverables:** [Deliverable](#), [Deliverable](#), [Deliverable](#)

**Summary & Research Context:** An annual measure of user satisfaction and pain points was collected from current Rasta (n=507). Customer satisfaction (CSAT) was 71.2 out of 100 and top-box was 78.7%. Top takeaways were that (1) Metric selection is hard without metric definitions and knowledge of what is relevant; (2) Custom slicing or condition is confusing without clear rules of slice crossing, finding the right slices, and association with metric display; and (3) Navigating to experiments is complicated for having to find experiment ID and figure out what is relevant.

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### 3. CPlan A/B Iteration-1 Study

- **Date / Status:** 2021-02-01 | **Team:** Data
- **Research Archive Link:** [CPlan A/B Iteration-1 Study](#)
- **Deliverables:** [Presso](#), [Report](#), [Summary](#)

**Summary & Research Context:** 4 participants participated in a 90-minute study over GVC to 1) Report on experiment costs, 2) Diagnose capacity issues, 3) Identify whether costs for experiments are acceptable, and 4) Find what the source of an experiment's impact was (e.g. device or traffic stream), followed by a 5-question survey about their experience after each task, follow-up interview questions, and a system usability survey at the end of the session. The participants evaluated both the old and the new version using the above mentioned tasks.

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### 4. Plx Explorer Iterative Test, 2019 Q4

- **Date / Status:** 2020-02-01 | **Team:** Data
- **Research Archive Link:** [Plx Explorer Iterative Test, 2019 Q4](#)
- **Deliverables:** [Deliverable](#), [Deliverable](#)

**Summary & Research Context:** The goal of this series of iterative testing is to get feedback on Plx Explorer so that the team can address usability issues and improve the user experiences before GA. Six participants (n = 6) participated in a 90-minute think-aloud over GVC using current Plx Explorer to do three tasks: (1) investigate the reasons behind bug spike; (2) discover the growth trends against different employee types; and (3) add a new field and describe the latency of resolving a bug. Participants filled out a 5-question survey about their experience after each task.

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### 5. Xenon 2019 Q4 CSAT Survey

- **Date / Status:** 2020-01-01 | **Team:** Data
- **Research Archive Link:** [Xenon 2019 Q4 CSAT Survey](#)
- **Deliverables:** [Xenon CSAT 2019 Q4 Results Summary](#), [Xenon 2019 Q4 CSAT Survey, Full Results Report](#), [Xenon 2019 Q4 CSAT Survey, Results Presentation](#)

**Summary & Research Context:** An annual measure of user satisfaction and pain points was collected from current Sawmill (n=116). Customer satisfaction (CSAT) was 76.1 out of 100 and top-box was 82.8%. Top takeaways were that (1) Insufficient error messages complicate troubleshooting; (2) Navigating the interface can be confusing or cumbersome; and (3) Compatibility with broader systems still feels limited or under supported to users.

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## 6. Q2 2022 Evals CSAT

- **Date / Status:** 2022-08-01 | **Team:** Data
- **Research Archive Link:** [Q2 2022 Evals CSAT](#)
- **Deliverables:** [Deliverable](#), [Deliverable](#), [Deliverable](#)

**Summary & Research Context:** The goal of this survey is to collect user satisfaction scores (CSAT), pain points, and identify priorities for Evals (+ Flower) in Q2 2022, in comparison with previous CSAT studies (Q2 2021). Of the 3439 users who received the survey, 406 participants completed (n=406) the survey. Our thematic analysis revealed that Evals instability and infrastructure issues, difficulty debugging and Evals configuration remained the most frustrating pain points. The CSAT average was approximately between Neutral and Slightly satisfied (M = 4.8) or 62.8 out of 100, confidence interval  $\pm 2.7$ . CSAT top-box (reports of any level of satisfaction) was 64.7%, with a margin of error  $\pm 4.3\%$ .

### Key Insights & Takeaways:

- Configuring parameters in Evals felt daunting and error prone in the lack of support in finding and understanding templates, options and the corresponding instructions
  - Debugging hurdles persist without actionable feedback, further increasing user frustration
  - Evals Instability & Infrastructure Issues Detrimentially Impact Reliability
- 

## 7. UGC Infrastructure Exploratory Study

- **Date / Status:** 2021-03-01 | **Team:** Data
- **Research Archive Link:** [UGC Infrastructure Exploratory Study](#)
- **Deliverables:** [report](#), [summary/highlight](#), [preso](#)

**Summary & Research Context:** The goal of this study was to reveal the Critical User Journeys (CUJs) of UGC Data Solutions (i.e. Zipit, ContribService, SBE and Geo UGCS) and uncover users' friction points. Nine participants were recruited for a 60-minute semi-structured interview via GVC. Open and axial coding to the transcribed data generated 2 takeaways, five CUJs and a range of friction points from the users' perspective. Leveraging these learnings and insights, the UGC team will focus on ways to simplify onboarding for customers including updating documentation, providing references and examples and aligning policies across the teams when integrating with other systems.

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## 8. Amarna & Raffia CSAT 2020 Q2

- **Date / Status:** 2020-08-01 | **Team:** Data
- **Research Archive Link:** [Amarna & Raffia CSAT 2020 Q2](#)
- **Deliverables:** [Preso](#), [Full Report](#), [summary/highlight](#)

**Summary & Research Context:** The study revealed the level of user satisfaction for Amarna is Moderately satisfied (M = 5.6) or 77.3±13.2 out of 100 on average, and that Raffia satisfaction is Moderately satisfied (M = 5.8) or 80.3±8.8 out of 100. Based on the revealed debuggability and system transparency issues, Hamilton team will continue working with UXR to dive deeper into the debuggability issues in a follow-up interview study, and refine the Hamilton Debuggability PRD.

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#### 9. ContribService (UGC Storage Infra) Usability Tests 2021 Q4 and 2022 Q1

- **Date / Status:** 2022-03-01 | **Team:** Data
- **Research Archive Link:** [ContribService \(UGC Storage Infra\) Usability Tests 2021 Q4 and 2022 Q1](#)
- **Deliverables:** [Full report](#), [summary](#), [Proposal and study design](#)

**Summary & Research Context:** After working through the usability issues identified through Round#1, Contribservice's average success rate for top critical user journeys increased from 58% (first study, on November 15, 2021) to 95% (last study on January 25, 2022); Average positive sentiment increased (+8 user sentiment points from 61.0 to 69.0 from round 1 to round 2). It signals the readiness of General Availability criteria in terms of CUJ velocity (95%), whereas further issues for more improvement were also identified in Round #2 of the Usability test to account for the varied usability scores.

#### Key Insights & Takeaways:

- ACL configuration is most challenging for ContribService users (6/8) because of the difficult-to-interpret jargons (e.g. "custom") and lack of examples.
  - Being able to copy and paste with a similar example in the codelab would make the task feel easy (5/8).
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#### 10. 2022 Q3 GoogleSQL User Segmentation Interview

- **Date / Status:** 2022-08-01 | **Team:** Data
- **Research Archive Link:** [2022 Q3 GoogleSQL User Segmentation Interview](#)
- **Deliverables:** [Deliverable](#), [Deliverable](#)

**Summary & Research Context:** Six user segments were qualitatively identified with distinctive goals, including 1) system developers, 2) system administrators, 3) data engineers, 4) data analysts, 5) investigators, and 6) general data consumers. System developers and administrators are essential in building the infrastructure for other user segments. Data engineers support part of the tasks of data analysts. Investigators are heavily dependent on dashboards to inform their

plans. The typical tasks for achieving goals for each user segment were also identified. There is a high degree of heterogeneity among each segment, with some overlap between data engineers, data analysts, and investigators and between system developers and system administrators.

### **Key Insights & Takeaways:**

- Emphasize the goals of system developer and system admin and reinforce their roles to support other segments
  - Challenges of system developers and system administrators: Writing/generating SQL involving reading repeated fields with cross join with a big dataset with PII.
  - Consider checking each segment and its goals to provide customized support from GoogleSQL to improve user satisfaction
  - Provide better support to the interaction (e.g., visualization and data storage tools) between different user segments to improve user satisfaction (use visualization as an example)
  - Challenges of data engineers, data analysts and investigators: monitor product status by creating a Plx dashboard [interaction] from Plx tables that are resulted from querying a BigQuery/Spanner dataset
  - Consider end-to-end support of data aggregation, analysis and visualization for data engineers, data analysts and investigators
- 

#### *11. literature review for data discovery*

- **Date / Status:** 2020-05-01 | **Team:** Data
- **Research Archive Link:** [literature review for data discovery](#)
- **Deliverables:** [Data Discovery UXR lit review \(go/discovery-sprint-lit-review\)](#)

**Summary & Research Context:** A literature review on research studies on why and how data workers (e.g. engineers, analysts) discover data revealed some occurring stages, such as identifying and scoping business needs, discovering and search for available datasets, making sense of datasets, evaluating datasets and leveraging datasets in product, model or analysis. Among these, involving experts has been a common yet time-consuming practice. The review also uncovered some challenges, including translating business needs into data related actions, lack of centralized data catalog and schemes, difficulty in finding and engaging with subject matter experts.

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#### *12. Plx Explorer Usability Test*

- **Date / Status:** 2020-07-01 | **Team:** Data
- **Research Archive Link:** [Plx Explorer Usability Test](#)
- **Deliverables:** [Deliverable](#), [Deliverable](#), [Deliverable](#)

**Summary & Research Context:** The goal of this usability test was to evaluate the usability of Plx Explorer CUJs, measuring a baseline for potential GA and future comparison. Nine participants (n = 9) participated in a 90-minute controlled laboratory test over GVC using Plx Explorer to do three tasks. Based on 5-question survey about their experience after each task, follow-up interviews and System Usability Score (SUS) at the end of the session, we found consistent usability issues of slicing and labels of [all value], friction points without autocomplete suggestions or matching strings in the typing process, and confusions caused by the inconsistent UI components.

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### *13. Data Assets Organization Study*

- **Date / Status:** 2022-03-01 | **Team:** Data
- **Research Archive Link:** [Data Assets Organization Study](#)
- **Deliverables:** [Full report](#), [Preso](#)

**Summary & Research Context:** Given Plx Home users' confusion with organizing assets and the concept of Collections, the interview study aims to understand Plx users' assets organization practices, why and how they fit into the data discovery or search processes, and to identify friction points and gaps in their data discovery, search and exploration experiences with Plx Home. In addition, the study results shed light on the five Critical User Journeys (CUJs) of Plx Home, with a focus on their motivations to visit the landing page so as to inform design directions of DDP collections and Plx homepage revamp.

### **Key Insights & Takeaways:**

- Surfacing assets related to users' interests (or interaction history) is pertinent while the default settings in My assets and Collections tabs are limiting and unintuitive (6/9).
  - Plx Home users leverage organizing tactics motivated by locating their own or others' assets, sharing with others, and curation for future use (9/9).
  - When search is hard or clueless, participants organize assets based on higher-level asset structure (5/9), asset types (3/9), and asset generating relations (3/9).
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### *14. Evals CSAT 2021 Q2*

- **Date / Status:** 2021-08-01 | **Team:** Data
- **Research Archive Link:** [Evals CSAT 2021 Q2](#)
- **Deliverables:** [Preso](#), [Summary archive in uxr group](#), [Full report](#)

**Summary & Research Context:** The survey measured user satisfaction and revealed their pain points and what worked well in 2021 Q2. Users who had run more than 5 Evals in the past 90 days were invited to participate in a Customer Satisfaction (CSAT) survey (n=554) between June

16-June 30, 2021 (23.2% response rate). Of the completed surveys, 389 respondents provided open-ended responses. At an Interrater Reliability Rate of 87.7%, our thematic analysis revealed that Evals instability associated with backend errors, infrastructure OMGs and CL sync issues remained the most frustrating pain point, which further exacerbated the difficulty of debugging. In an independent-samples Welch's t-test of satisfaction responses comparing 2020 Q4 to 2021 Q2, there was a significant increase from 2020 (Topbox=62.2%, M=60.9, SD=29.7) to 2021 (Topbox=72.0%, M=65.6, SD=27.8),  $t(1200)=-2.86$ ,  $p = 0.04$ .

### Key Insights & Takeaways:

- Evals instability ascribed to backend errors, infrastructure OMGs and CL sync issues (186/389) exacerbated the difficulty of debugging (106/389).
  - Configuring parameters in Evals felt daunting and error prone in the lack of support in finding and understanding templates, options and the corresponding instructions (60/389).
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#### 15. Evals CSAT 2020 Q2

- **Date / Status:** 2020-08-01 | **Team:** Data
- **Research Archive Link:** [Evals CSAT 2020 Q2](#)
- **Deliverables:** [Full Report](#), [proposal](#), [preso](#)

**Summary & Research Context:** A customer satisfaction survey that consisted of likert scale ratings of Evals satisfaction and open-ended responses was distributed to active Eval users over the past 90 days in 2020 Q2 (n = 515). The average satisfaction of Evals was 59.2/100  $\pm$  2.6 (top box, or any level of satisfaction ratings, 60.8%  $\pm$  4.1%), indicating a significant increase (+6.8/100) of satisfaction score compared with the previous satisfaction score collected 6 months ago. It further revealed users' painpoints around reliability, configuration, debugging and latency.

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#### 16. gORM C++ Feature Prioritization with MaxDiff Survey

- **Date / Status:** 2020-05-01 | **Team:** Data
- **Research Archive Link:** [gORM C++ Feature Prioritization with MaxDiff Survey](#)
- **Deliverables:** [Slides](#), [Report](#), [Summary](#)

**Summary & Research Context:** To inform the product feature prioritization for a gORM-like (storage object mapping) tool in C++ against the existing Java offering, we conducted a MaxDiff survey to assess user interests among 12 features prepared by the gORM team (n = 108). The results indicate a clear interest overall in a gORM offering for C++ and provide insights to prioritize features in future implementations with 4 buckets of feature groups.



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### 17. Indexing Solution CUJ Exploration

- **Date / Status:** 2022-01-01 | **Team:** Data
- **Research Archive Link:** [Indexing Solution CUJ Exploration](#)
- **Deliverables:** [Full report](#), [slides](#), [summary](#)

**Summary & Research Context:** The study aims to explore how developers built and iterated on indexing pipelines at Google through various layers of infrastructure, components and services. The study uncovered 5 Critical User Journeys (CUJs) involved in the process of building, monitoring, and debugging indexing pipelines, based on GVC interviews for a total duration of over 9 hours with eight participants (n = 8). Their interaction with the indexing pipeline varied across the starting point, product adoption and engagement level, signal (or data) ownership and data types, but revealed a common set of critical user journeys that cover multiple stages. This exploratory study also shed light upon how indexing pipeline users suffered from the lack of monitoring and alerting support, and pointed out pain points of using debugging toolings at both the URL level and batch level.

#### Key Insights & Takeaways:

- Indexing Solutions users suffered from fragmented indexing services that span a wide spectrum of stages and hard-to-use debug information and toolings (6/8).
- The study aims to explore how developers built and iterated on indexing pipelines at Google through various layers of infrastructure, components and services. The study uncovered 5 Critical User Journeys (CUJs) involved in the process of building, monitoring, and debugging indexing pipelines, based on GVC interviews for a total duration of over 9 hours with eight participants (n = 8). Their interaction with the indexing pipeline varied across the starting point, product adoption and engagement level, signal (or data) ownership and data types, but revealed a common set of critical user journeys that cover multiple stages.
- Indexing Solutions users lack alerting instrumentation and monitoring mechanisms (7/8).

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### 18. Data Discovery Design Sprint

- **Date / Status:** 2020-06-01 | **Team:** Data
- **Research Archive Link:** [Data Discovery Design Sprint](#)
- **Deliverables:** [design sprint review deck](#), [design sprint activity deck](#)

**Summary & Research Context:** The four-day design sprint reviewed past work through several talks, and proposed a new product vision to make data discovery easier for Googlers.

#### Key Insights & Takeaways:

- Data Discovery should expose assets relevant to the point of interests and action
  - It is crucial to promote interactivity on the data discovery journey by engaging users both as information consumers and also as data information contributors
  - It is crucial to align with existing user behaviors, such as schema, sample data and annotation while onboarding new forms (data card, network view) with educational materials to cater to the learning curve
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